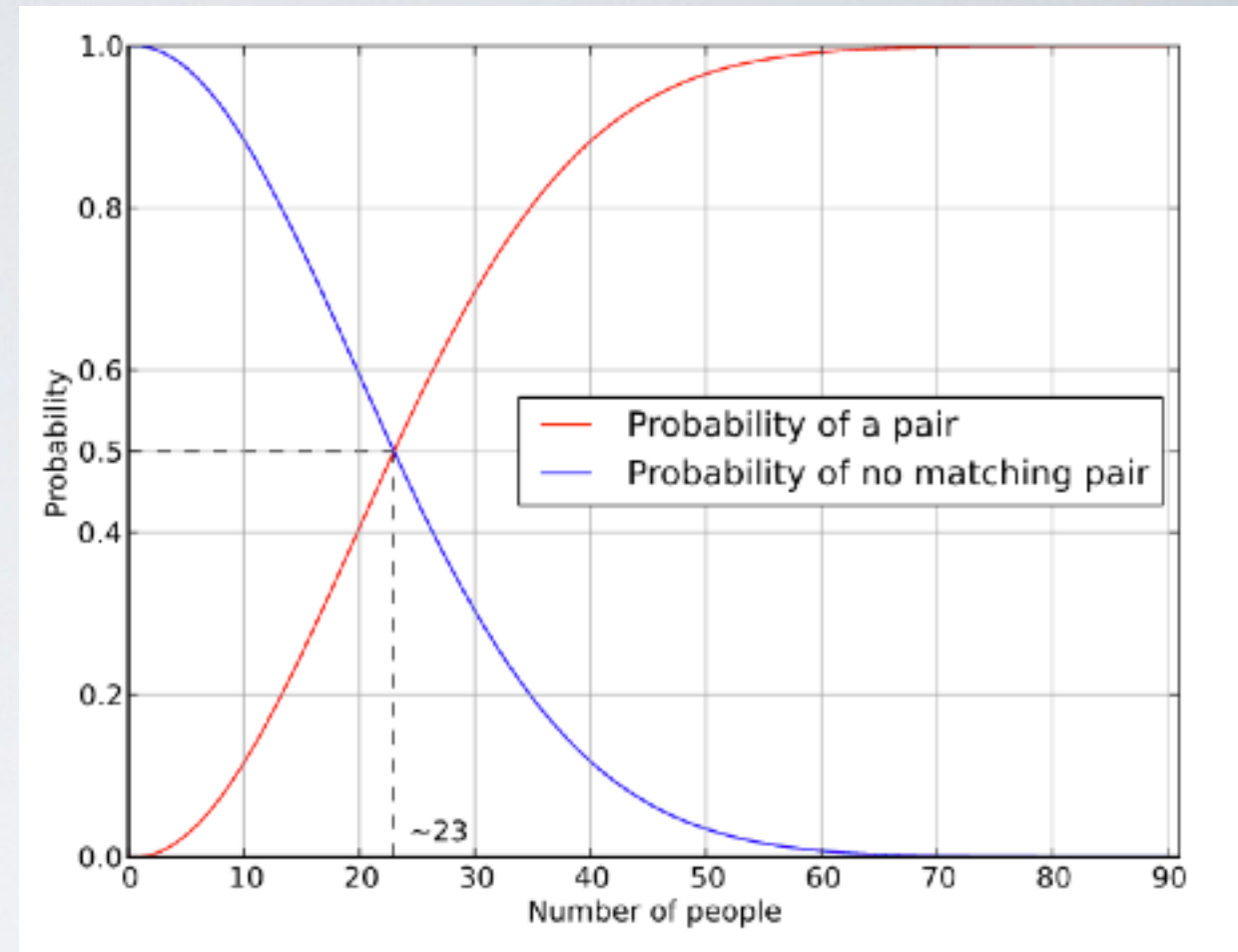


Birthday Paradox

“There are 50% chance that 2 people have the same birthday in a room of 23 people”



N-bits security

- ➡ Given a hash function **H** of **n** bits output, a collision can be found in around **$2^{n/2}$** evaluations
e.g SHA-256 is 128 bits security

Broken hash functions beyond the birthday paradox

	Year	Collision
MD5	2013	2^{24} evaluations (2^{39} with prefix)
SHA-1	2015	2^{57} evaluations